Ratios

A ratio compares two amounts: for every 3 of these, 2 of those.

3 circles for every 2 squares



3 : 2

Say “3 to 2”. Two groups is 6 : 4, the same ratio.

A ratio table: multiply both rows by the same number

cups of flour	2	4	6	8
eggs	3	6	9	12

× 2, × 3, × 4:
both numbers change together.

Part to part, part to whole

A class has 3 boys for every 5 girls. Boys to girls is 3 : 5.
Out of every 8 kids, 3 are boys. Boys to all kids is 3 : 8.

Your turn!

$$4 : 5 = 12 : \square$$

Ratios: Practice

Both numbers change together.

Part 1 Write the ratio Count the circles, then the squares.

1) ●●● □□

circles to squares = ☐ : ☐

2) ●●●● □

circles to squares = ☐ : ☐

3) ●● □□□□□

circles to squares = ☐ : ☐

4) ●●●●● □□□

circles to squares = ☐ : ☐

5) ● □□□□

circles to squares = ☐ : ☐

6) ●●●●●● □□

circles to squares = ☐ : ☐

7) ●●● □□□□

circles to squares = ☐ : ☐

8) ●● □□□

circles to squares = ☐ : ☐

9) ●●●●● □□

circles to squares = ☐ : ☐

Part 2 Ratio tables Multiply both rows by the same number.

10)

cups	2	☐	☐	☐	☐
people	3	☐	☐	☐	☐

11)

laps	3	☐	☐	☐	☐
minutes	4	☐	☐	☐	☐

12)

red	5	☐	☐	☐	☐
blue	2	☐	☐	☐	☐

Ratios: Practice (continued)

Part 2 Ratio tables (continued)

13)

pens	4				
dollars	3				

14)

cats	1				
dogs	6				

15)

boys	7				
girls	5				

Part 3 Equal ratios No help this time.

16) $2 : 3 = \square : 12$

17) $5 : 4 = 15 : \square$

18) $1 : 6 = \square : 30$

19) $3 : 7 = 6 : \square$

20) $4 : 9 = \square : 27$

21) $6 : 5 = 24 : \square$

22) $2 : 9 = \square : 54$

23) $7 : 3 = 35 : \square$

24) $8 : 5 = \square : 10$

Ratios: Practice (continued)

Part 3 Equal ratios (continued)

25) $3 : 10 = 12 : \square$

26) $9 : 4 = \square : 12$

27) $5 : 8 = 30 : \square$

28) $4 : 7 = \square : 35$

29) $1 : 9 = 8 : \square$

30) $6 : 7 = \square : 21$

31) $10 : 3 = 20 : \square$

32) $3 : 8 = \square : 56$

33) $2 : 11 = 6 : \square$

34) $7 : 4 = \square : 24$

35) $5 : 9 = 20 : \square$

36) $8 : 3 = \square : 9$

Ratios: Practice (continued)

Part 3 Equal ratios (continued)

37) $11 : 4 = 22 : \square$

38) $3 : 5 = \square : 45$

39) $12 : 5 = 24 : \square$

Part 4 Simplest form Divide both numbers by the same number.

40) $8 : 12 = \square : \square$

41) $6 : 10 = \square : \square$

42) $6 : 24 = \square : \square$

43) $15 : 6 = \square : \square$

44) $8 : 14 = \square : \square$

45) $9 : 24 = \square : \square$

46) $28 : 8 = \square : \square$

47) $25 : 30 = \square : \square$

48) $6 : 27 = \square : \square$

Ratios: Practice (continued)

Part 4 Simplest form (continued)

49) 12 : 10 = ☐ : ☐

50) 7 : 21 = ☐ : ☐

51) 16 : 6 = ☐ : ☐

Part 5 Part to whole No help this time.

52) 3 boys for every 5 girls: boys to all kids = ☐ : ☐

53) 2 boys for every 7 girls: boys to all kids = ☐ : ☐

54) 4 boys for every 3 girls: boys to all kids = ☐ : ☐

55) 1 boy for every 5 girls: boys to all kids = ☐ : ☐

56) 5 boys for every 6 girls: boys to all kids = ☐ : ☐

57) 6 boys for every 1 girl: boys to all kids = ☐ : ☐

58) 2 boys for every 3 girls: boys to all kids = ☐ : ☐

59) 7 boys for every 4 girls: boys to all kids = ☐ : ☐

Ratios: Practice (continued)

Part 6 More equal ratios

60) $2 : 6 = 4 : \square$

61) $4 : 8 = \square : 32$

62) $2 : 4 = \square : 36$

63) $11 : 12 = \square : 84$

64) $3 : 5 = 18 : \square$

65) $11 : 7 = 77 : \square$

66) $7 : 12 = \square : 84$

67) $3 : 12 = \square : 36$

68) $1 : 2 = 7 : \square$

69) $6 : 4 = 48 : \square$

70) $12 : 11 = \square : 55$

71) $6 : 12 = 24 : \square$

Ratios: Practice (continued)

Part 6 More equal ratios (continued)

72) $9 : 3 = 18 : \square$

73) $10 : 12 = \square : 24$

74) $9 : 4 = 81 : \square$

75) $10 : 2 = 30 : \square$

76) $2 : 7 = \square : 28$

77) $10 : 6 = 20 : \square$

78) $5 : 8 = \square : 64$

79) $2 : 5 = 12 : \square$

80) $4 : 3 = \square : 15$

81) $11 : 7 = \square : 56$

82) $6 : 3 = 36 : \square$

83) $12 : 10 = \square : 30$

Ratios: Practice (continued)

Part 7 More simplest form

84) $12 : 32 = \square : \square$

85) $3 : 18 = \square : \square$

86) $4 : 10 = \square : \square$

87) $35 : 40 = \square : \square$

88) $27 : 6 = \square : \square$

89) $14 : 49 = \square : \square$

90) $25 : 15 = \square : \square$

91) $27 : 12 = \square : \square$

92) $32 : 20 = \square : \square$

93) $2 : 4 = \square : \square$

94) $6 : 36 = \square : \square$

95) $4 : 6 = \square : \square$

Ratios: Practice (continued)

Part 8 Remember? Problems from earlier packets.

96) $(6 + 9) \times 4 = \square$

97) $9 \times \frac{2}{7} = \frac{\square}{7}$

98) $0.6 \times 0.9 = \square$

99) $3{,}472 \div 8 = \square$

100) $\frac{7}{8} - \frac{1}{3} = \frac{\square}{\square}$

101) $4 \text{ ft } 2 \text{ in} = \square \text{ in}$

Part 9 Word problems Use a ratio table if it helps.

102) A recipe uses 2 cups of rice for every 3 cups of water. How much water goes with 8 cups of rice?

Number sentence _____

Answer ☐ cups

Draw or show your work

103) There are 4 red marbles for every 5 blue ones. There are 20 blue marbles. How many red ones?

Number sentence _____

Answer ☐ red marbles

Draw or show your work

104) A team won 6 games and lost 9. What is the ratio of wins to losses in simplest form?

Number sentence _____

Answer ☐

Draw or show your work

Ratios: Practice (continued)

Part 9 Word problems (continued)

105) Paint is mixed 3 cans of blue to 1 can of white. How many cans of blue go with 7 cans of white?

Number sentence _____

Answer ☐ cans

Draw or show your work

106) Challenge: A class has 3 boys for every 4 girls. There are 28 kids. How many are girls?

Number sentence _____

Answer ☐ girls

Draw or show your work

Check-In: Ratios

Show what you know. No help this time. Score: _____ / 7

Part 1 Write the ratio

1) 

circles to squares = ☐ : ☐

Part 2 Ratio tables

2)

cups	3	☐	☐	☐
people	5	☐	☐	☐

Part 3 Equal ratios

3) $3 : 4 = \square : 20$

4) $4 : 3 = 24 : \square$

5) $5 : 7 = 10 : \square$

Part 4 Simplest form

6) $8 : 28 = \square : \square$

7) $10 : 6 = \square : \square$